Contractor Termination Activity

AccessGuard Health – Healthcare IAM Operations Simulation

Leaver scenario: fully deprovisioning a contractor and proving he can no longer get in

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Activity	Activity 3 – Contractor Termination (Leaver)
Scope	On-premises directory (Samba4 Active Directory)
Document Version	1.0
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1. Purpose

The **leaver** is the last step of the identity lifecycle: when someone's engagement ends, every way in must be closed. A contractor whose account stays active after he leaves is one of the most common real-world risks. This activity terminates a contractor on the on-premises directory: the account is disabled, his group memberships are removed, and a test from the client proves he can no longer sign in.

The employee is fictional. In Microsoft Entra ID the same leaver was recorded earlier (ticket HVMC-JML-203); this activity applies it to the on-premises directory so both agree.

2. Scenario

Item	Before	After
Employee	Carlos Diaz (fictional), contractor	Carlos Diaz, terminated
Username	carlos.diaz	carlos.diaz
OU	OU=Contractors,OU=Users	OU=Contractors,OU=Users (kept for audit)
Account state	Enabled (userAccountControl 512)	Disabled (userAccountControl 514)
Group	SG-Contractors-VPN-Read	None
Access	Read-only on the VPN and ITAdmin shares	None

On-premises, application access comes only from group membership (the Contractors group is read-only on the VPN and ITAdmin shares), so removing the group removes all of his app access. The account is **disabled, not deleted**, so the record stays available for audit.

3. Step 1 – Find the Contractor

The only member of the Contractors group was listed:

```
sudo samba-tool group listmembers SG-Contractors-VPN-Read
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool group listmembers SG-Contractors-VPN-Read
[sudo] password for ubuntu:
carlos.diaz
ubuntu@hvmc-dc01:~$
```

Figure 1: The contractor is carlos.diaz

4. Step 2 – Before: Account State and Group

His location, account state and group were recorded before any change. A `userAccountControl` value of 512 means the account is enabled.

```
sudo samba-tool user show carlos.diaz | grep -E "^dn:|memberOf|userAccountControl"
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user show carlos.diaz | grep -E "^dn:|memberOf|userAccountControl"
dn: CN=Carlos Diaz,OU=Contractors,OU=Users,DC=hvmc,DC=lan
userAccountControl: 512
memberOf: CN=SG-Contractors-VPN-Read,OU=Groups,DC=hvmc,DC=lan
ubuntu@hvmc-dc01:~$
```

Figure 2: Before: enabled (512), in OU=Contractors, member of SG-Contractors-VPN-Read

5. Step 3 – Set a Lab Password for Testing

A lab password was set so the before and after tests could be run as Carlos. It follows the domain policy and is not shown in any screenshot.

```
sudo samba-tool user setpassword carlos.diaz
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user setpassword carlos.diaz
New Password:
Retype Password:
Changed password OK
ubuntu@hvmc-dc01:~$
```

Figure 3: Password changed

6. Step 4 – Before-Test: Open the VPN Share

From the client **hvmc-client01**, Carlos connected to the VPN share, which his group allows him to read:

```
smbclient //hvmc-dc01.hvmc.lan/VPN -U 'HVMC\carlos.diaz' -c 'ls'
```

```
ubuntu@hvmc-client01:~$ smbclient //hvmc-dc01.hvmc.lan/VPN -U 'HVMC\carlos.diaz' -c 'ls'
Password for [HVMC\carlos.diaz]:
  .                                   D        0  Tue Nov  3 04:12:54 2026
  ..                                  D        0  Tue Nov  3 04:12:54 2026

                10218772 blocks of size 1024. 4787740 blocks available
ubuntu@hvmc-client01:~$
```

Figure 4: Before: Carlos signs in and lists the VPN share (allowed)

7. Step 5 – The Termination

Two commands were run on the domain controller, in this order. The account is **disabled first**, which stops any new sign-in at once, and then the group is removed.

```
sudo samba-tool user disable carlos.diaz
sudo samba-tool group removemembers SG-Contractors-VPN-Read carlos.diaz
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user disable carlos.diaz
ubuntu@hvmc-dc01:~$ sudo samba-tool group removemembers SG-Contractors-VPN-Read carlos.diaz
Removed members from group SG-Contractors-VPN-Read
ubuntu@hvmc-dc01:~$
```

Figure 5: The account is disabled and the group membership removed

8. Step 6 – After: Account State and Group

The same check as Step 2 was repeated:

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user show carlos.diaz | grep -E "^dn:|memberOf|userAccountControl"
dn: CN=Carlos Diaz,OU=Contractors,OU=Users,DC=hvmc,DC=lan
userAccountControl: 514
ubuntu@hvmc-dc01:~$
```

Figure 6: After: disabled (514), no group, account kept in OU=Contractors

9. Step 7 – After-Test: Sign-In Is Refused

The before-test was repeated. The password is still the right one, but the account is disabled:

```
smbclient //hvmc-dc01.hvmc.lan/VPN -U 'HVMC\carlos.diaz' -c 'ls'
```

```
ubuntu@hvmc-client01:~$ smbclient //hvmc-dc01.hvmc.lan/VPN -U 'HVMC\carlos.diaz' -c 'ls'
Password for [HVMC\carlos.diaz]:
session setup failed: NT_STATUS_ACCOUNT_DISABLED
ubuntu@hvmc-client01:~$
```

Figure 7: After: sign-in fails with NT_STATUS_ACCOUNT_DISABLED, so no share can be opened

The error is different from the access-denied error seen in the onboarding and transfer tests. `NT_STATUS_ACCESS_DENIED` means the user signed in but is not allowed into a folder; `NT_STATUS_ACCOUNT_DISABLED` means the account itself is switched off, so the user cannot sign in at all.

10. Results

Check	Expected	Result
Account state before	Enabled	512 (enabled)
Group before	SG-Contractors-VPN-Read	SG-Contractors-VPN-Read
Open VPN share before	Allowed	Allowed (folder listed)
Account state after	Disabled	514 (disabled)
Group after	None	None
Sign in after	Refused	Refused: NT_STATUS_ACCOUNT_DISABLED
Account record	Kept for audit	Kept in OU=Contractors

11. Troubleshooting Log

Issue	Cause	Resolution
Error "Unable to open tdb /var/lib/samba/private/sam.ldb"	The disable command was run on the client machine, which holds no directory database	Nothing was changed; the command was repeated on the domain controller

12. Security Considerations

- All names and users are **fictional**; no real contractor or patient data is used.
- **Disable first:** the account is switched off before the clean-up, so there is no gap in which he could still sign in.
- **Disabled, not deleted:** keeping the account gives an audit trail and keeps its history.
- **Access through groups:** removing the group removed every app permission at once.
- A terminated user's old password stays useless: a disabled account cannot sign in even with the correct password.
- The lab password is practice-only and does not appear in the screenshots.

13. Key Outcomes

- Fully deprovisioned a contractor on the on-premises directory, matching the Entra ID record (HVMC-JML-203).
- Showed the before and after state of the account and its group, with evidence at each step.
- Proved from a real client that the terminated account can no longer sign in.

14. Deliverables

- Contractor Termination Activity: Lab Documentation for GitHub Portfolio (this document).
- Screenshots: before and after account state and group, the termination commands, and the before and after sign-in tests.